UNIVERSITY OF PENNSYLVANIA ESE 5700

Project 1, Part 3: the MAC unit

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Cadence GPDK 18 nm $V_{DD} = 0.8$ V, nominal corner, 27°C $T_{CLK,\min}$ = **[TODO: ns]** 17 cycles START→DONE
(predicted) area ≈ 292 μm^2 *(est.)*

Rubric map (handout, Part 3: "Report" list and 5-point grading)

Required item	Pts	Section	Status
Register cell with its characterization (setup, hold, t_{cq}, C_{CLK}, energy per clock)	1	§2	**Missing**: cell designed and fixed; `tb_dff` not run
Multiplier and adder schematics with sizes; logical-effort sizing of the critical path	1.5	§3	**Partial**: multiplier verified (256 pairs), worst case ≈ 1.2 ns measured, schematics in; logical effort missing
State diagram, sequencer schematic, one-dot-product timing diagram	1.5 (with timing closure)	§4	**Partial**: state diagram and logic described from the netlist; schematics, benches, timing diagram missing
Timing closure: hand-computed critical path vs. measured $T_{CLK,\min}$; hold check	(above)	§5	**Missing**: critical path named; numbers need §2 and `tb_top`
Verification waveforms (Test A) and pass/fail table	1 (with metrics and report quality)	§6	**Missing**: `tb_top` not built
Metrics table: FOM$_{MAC}$, D, V_{DD}; energy per MAC with and without memory; where the energy goes	(above)	§7	**Partial**: area estimated; delay, energy, FOM missing
Short paragraph: what limited the design		§8	**Partial**: first draft; final numbers missing
Top-level block diagram and schematic; team members and contributions		§1	**Partial**: `top` schematic in; block diagram and contributions missing

Update the status colors (red missing, yellow partial, green done) as sections fill in. All simulations are schematic level (no extracted parasitics): bitline capacitance $C_{BL} = 3.5$ fF per bitline (`cbw`, as in Part 2), and each block output under test drives one `inv_2x`. Strategy from the handout: treat the SRAM as a black box with the Part 2 interface and t_{access}; register first, then the multiplier (exhaustive, all 256 pairs), then the accumulator, then the sequencer; simulate the full unit only when every piece passes alone.

1. Team members and contributions

Member	Contribution
Victor Wanjohi	**[TODO: fill in]**
Tyrone Marhguy	**[TODO: fill in]**

Summary. Two copies of the Part 2 macro hold the weights W and activations X; a static-CMOS array multiplier, a 12-bit ripple-carry accumulator, a 4-bit address counter and a two-bit controller (all registers are transmission-gate

master-slave flip-flops) compute $\mathrm{ACC} = \sum_{i=0}^{15} W[i]\,X[i]$. The read is pipelined against the multiply-accumulate (read in cycle n, accumulate in $n+1$), so a dot product takes 17 cycles from the START edge to DONE (cycle-level model of the netlist). The multiplier is correct for all 256 input pairs and its worst case is about 1.2 ns (1.14–1.24 ns), so it, not the 488 ps SRAM read, sets the clock period. Area $\approx 292\ \mu\mathrm{m}^2$ *(est.)*. **[TODO: $T_{CLK,\min}$, Tests A/B/C result, energy.]**

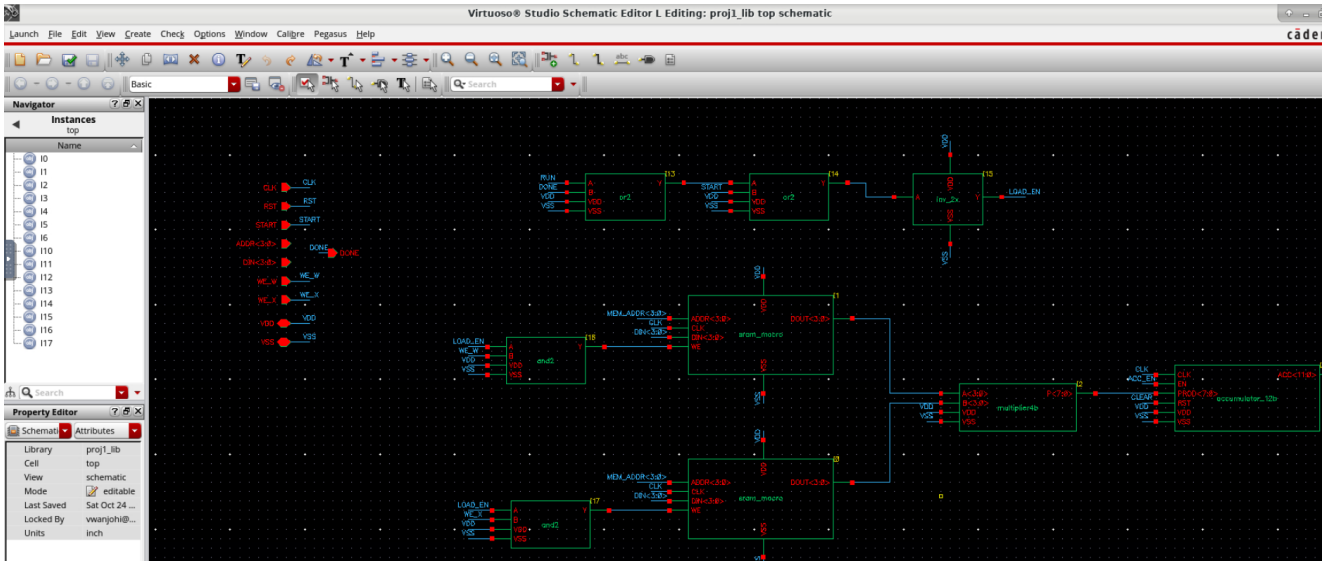


Figure 1: `top`, left half: the two `sram_macro` instances (I1 weights, written when LOAD_EN·WE_W; I0 activations, LOAD_EN·WE_X), sharing MEM_ADDR, CLK and DIN; their DOUT buses feed `multiplier4b` (I2), whose product feeds `accumulator_12b` (I3). Top: $\mathrm{LOAD_EN} = \overline{\mathrm{RUN} + \mathrm{DONE} + \mathrm{START}}$ (two `or2`, `inv_2x`).

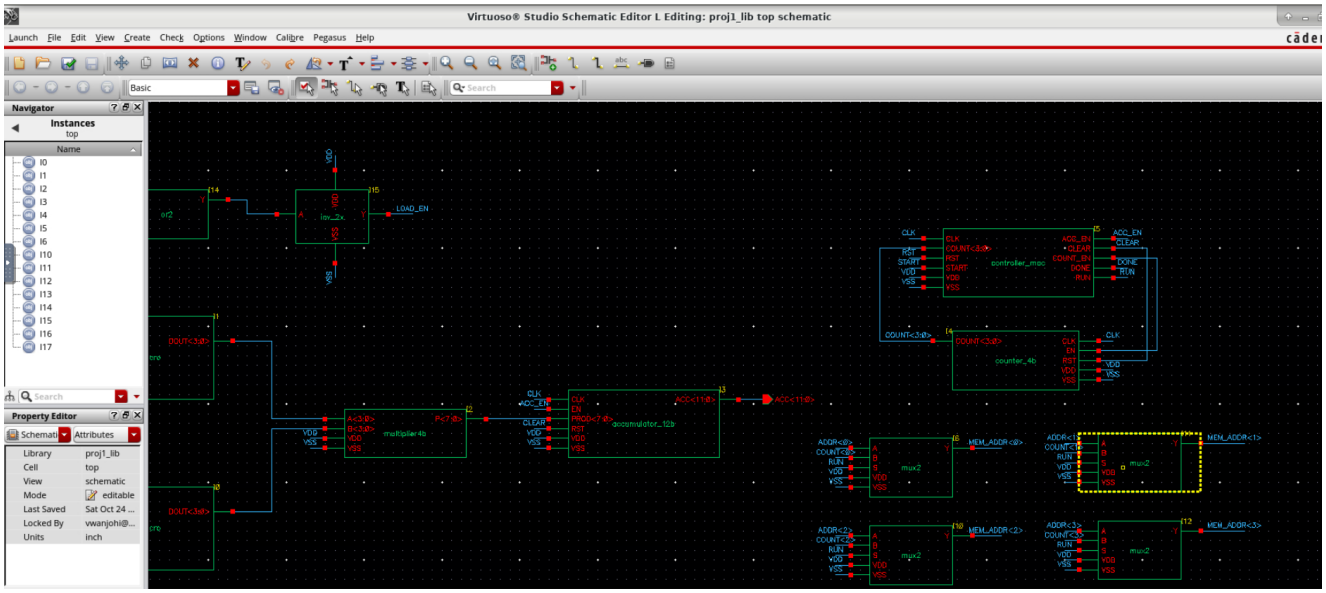


Figure 2: `top`, right half: `controller_mac` (I5) and `counter_4b` (I4) in a loop (COUNT_EN, CLEAR, COUNT⟨3:0⟩), and four `mux2` that send COUNT to MEM_ADDR while RUN = 1 and the external ADDR otherwise.

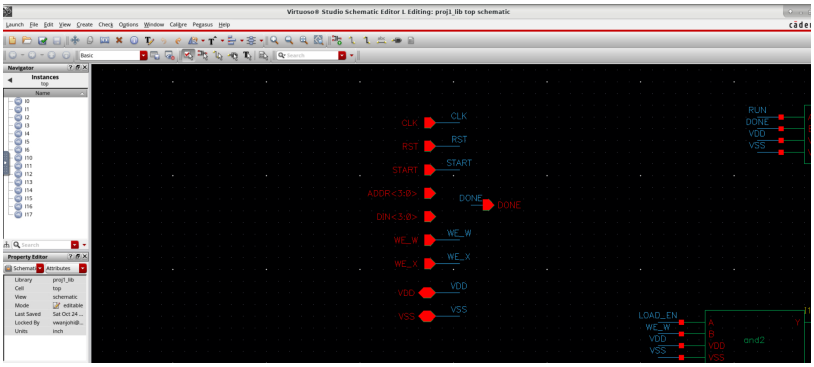


Figure 3: `top` ports: CLK, RST, START, ADDR⟨3:0⟩, DIN⟨3:0⟩, WE_W, WE_X in; DONE and ACC⟨11:0⟩ out; VDD, VSS.

Block inventory and test benches.

Block	Built from	Test bench	Status
Register bit (DFF)	latch pair, T-gates	`tb_dff`	designed; hold-loop bug fixed; not simulated
4×4 multiplier	AND array, adders	`tb_multiplier4b`	256/256 correct; worst case 1.14–1.24 ns
12-bit accumulator	adder 12b, 12 registers	`tb_accu_12b`	not simulated
4-bit counter	4 DFFs, toggle chain	`tb_counter4b`	stimulus and expected values ready
Controller (idle/compute/done)	4 registers, gates	`tb_controller`	stimulus and expected values ready
MAC top	all of the above and 2 macros	`tb_top`	not built

2. Sequential elements (3.1)

Rubric (3.1): register bit at transistor level, used for accumulator, counter, controller state, DONE; setup, hold, clock-to-Q with the simulation that determines each; clock-input capacitance; energy per clock; style justified.

Style and justification. Positive-edge master-slave flip-flop built from two static transmission-gate latches (`dlatch`): the master is transparent while CLK = 0, the slave while CLK = 1, so D is captured on the rising edge. Each latch has an input `tg`, a storage node driven by two `inv_2x`, a hold `tg` that closes the loop from Q, and its own clock inverter; the flip-flop adds one more `inv_2x` to make the master's clock (100 fins: `tg` 2+2 fins, `inv_2x` 8/4 fins). It was chosen because it reuses the verified `tg` and inverter cells, is fully static (holds with the clock stopped, unlike a dynamic C^{2}MOS latch), and needs only the single external clock. Its cost is the local clock inverters (clock load, and a small skew between the two phases, which hold must cover). `register_1b` adds a `mux2` load enable ($EN = 1$ loads D, $EN = 0$ recirculates Q) and an `and2` synchronous reset (152 fins). **[TODO: Alternative considered (e.g. C^{2}MOS or pulsed latch) and why rejected, with numbers once `tb_dff` has run.]**

Design note. The first `dlatch` fed the node after one inversion back through the hold gate, an inverting loop that cannot store a value; the hold `tg` now returns Q (two inversions) to the storage node and conducts while CLK = 0 (fixed 10/25, before any register simulation).

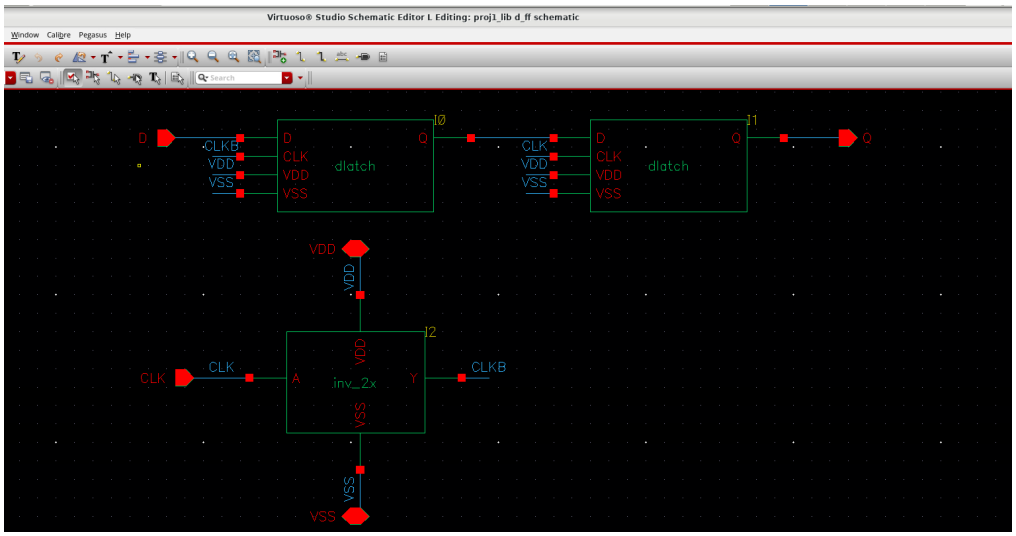


Figure 4: `d_ff`: master `dlatch` I0 clocked by CLKB (transparent while CLK = 0), slave I1 clocked by CLK, and the clock inverter I2 (`inv_2x`, 8/4 fins). Each `dlatch`: input and hold `tg` (2+2 fins), two `inv_2x` in the loop and one for its local clock; 100 fins in total.

Test bench (`tb_dff`): **[TODO: D swept relative to the CLK edge, loaded by one register input or FO4, realistic input slews. Describe the bisection or sweep.]**

Quantity	Value	Determined by
Setup time t_{su} (D 0→1 / 1→0)	**[TODO: ps]**	**[TODO: D-to-CLK offset at which Q fails, or CLK-to-Q rises by 10% (state the criterion)]**
Hold time t_h	**[TODO: ps]**	**[TODO: same criterion, D changing after the edge]**
Clock-to-Q t_{cq} (rise / fall)	**[TODO: ps]**	**[TODO: CLK 50% to Q 50%, D stable well before the edge]**
Clock-input capacitance C_{CLK}	**[TODO: fF]**	**[TODO: charge per cycle from CLK source / V_{DD}, or AC method]**
Energy per clock	**[TODO: fJ]**	**[TODO: integral of V_{DD} current over one cycle, with D toggling / not toggling]**

[missing figure: tb_dff_setup_sweep]

Figure 5: **[TODO: Setup-time determination: t_{cq} against D-to-CLK offset, showing the wall.]**

[missing figure: tb_dff_hold_sweep]

Figure 6: **[TODO: Hold-time determination.]**

[missing figure: tb_dff_tran]

Figure 7: **[TODO: Clock-to-Q transient and the energy measurement window.]**

3. Datapath (3.2)

Rubric (3.2): 4×4 unsigned multiplier and 12-bit accumulator adder; any architecture justified; critical path sized with logical effort before simulating.

Leaf cells. Static CMOS throughout, $L = 18$ nm: `tg` (2+2 fins), `mux2` (two `tg` and an `inv_2x`, $Y = S\,?\,B : A$), `xor2` (complementary static gate with input inverters, 44 fins), `and2` (`nand2_u`+`inv_2x`), `or2` (NOR2+`inv_2x`). The full adder is $S = (A \oplus B) \oplus C_{in}$, $C_{out} = C_{in}(A \oplus B) + AB$ from two `xor2`, two `and2` and an `or2` (148 fins).

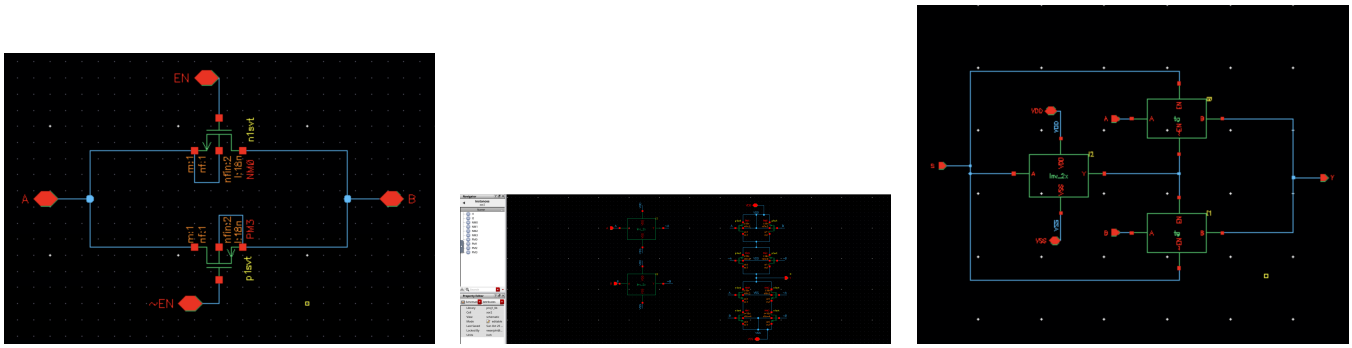


Figure 8: Leaf cells: `tg` (nfin 2 each), `xor2`, `mux2`. The `xor2` view is the corrected one (two series pull-down stacks). **[TODO: Re-capture `mux2` after the select swap.]**

Multiplier. Array multiplier: 16 AND gates form the partial products $a_i b_j$ (four `and2x4`, one per row j), and three 4-bit ripple-carry adders (`fulladder_4b`, 12 full adders) sum the rows, each row shifted by one bit; $P_0 = a_0 b_0$, $P_{1..3}$ from the low bit of each adder row, $P_{4..7}$ from the last row and its carry. It reuses one verified full-adder cell in a regular array, which made it quick to build and debug; the cost is speed, because the longest path ripples through all three adder rows into the last row's carry (P_6, P_7). A carry-save (Wallace or Dadda) tree would shorten the path to two full-adder levels plus one final carry-propagate adder, for the same 12 full adders but irregular wiring. 2096 fins.

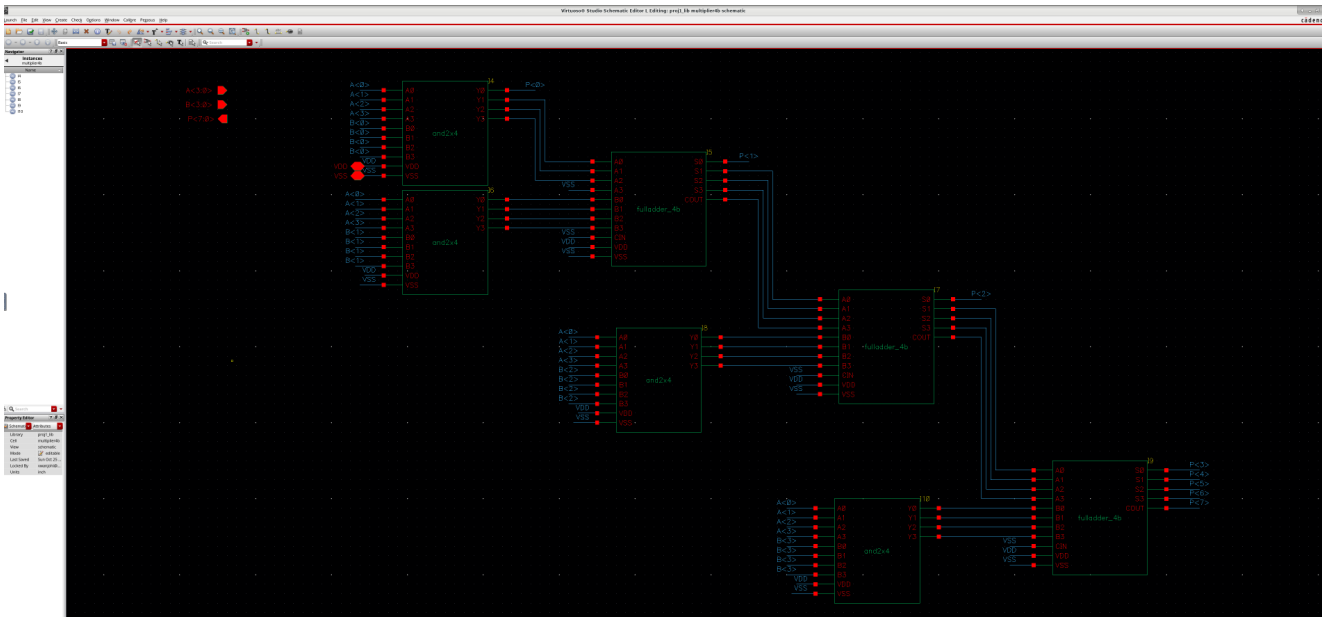


Figure 9: `multiplier4b` (corrected): partial-product rows I4, I6, I8, I10 (`and2x4`, row j gets $A\langle 0..3\rangle$ and B_j on all four B pins) and ripple rows I5, I7, I9 (`fulladder_4b`, $C_{in} = 0$); each row adds the previous row's upper sum bits and carry to the next partial product. Sizes ($L = 18$ nm): `and2` = `nand2_u` (PMOS 2‖2, NMOS 2+2 series fins) + `inv_2x` (8/4); full adder: two `xor2` (NMOS 2/4, PMOS 2 fins, input `inv_2x`), two `and2`, one `or2`.

Accumulator adder. 12-bit ripple-carry adder of the same full adder (`fulladder_12b`): $A = \mathrm{ACC}\langle 11{:}0\rangle$, $B = \{0000, P\langle 7{:}0\rangle\}$, $C_{in} = 0$; the sum goes to `register_12b` (12 `register_1b`, $EN = \mathrm{ACC_EN}$, synchronous reset on

CLEAR). Twelve bits are enough: the largest result is $16 \times 15 \times 15 = 3600 < 4096$. Only the low 8 bits receive the product, so the upper 4 stages only propagate the carry. 3600 fins for the accumulator (1776 adder, 1824 register).

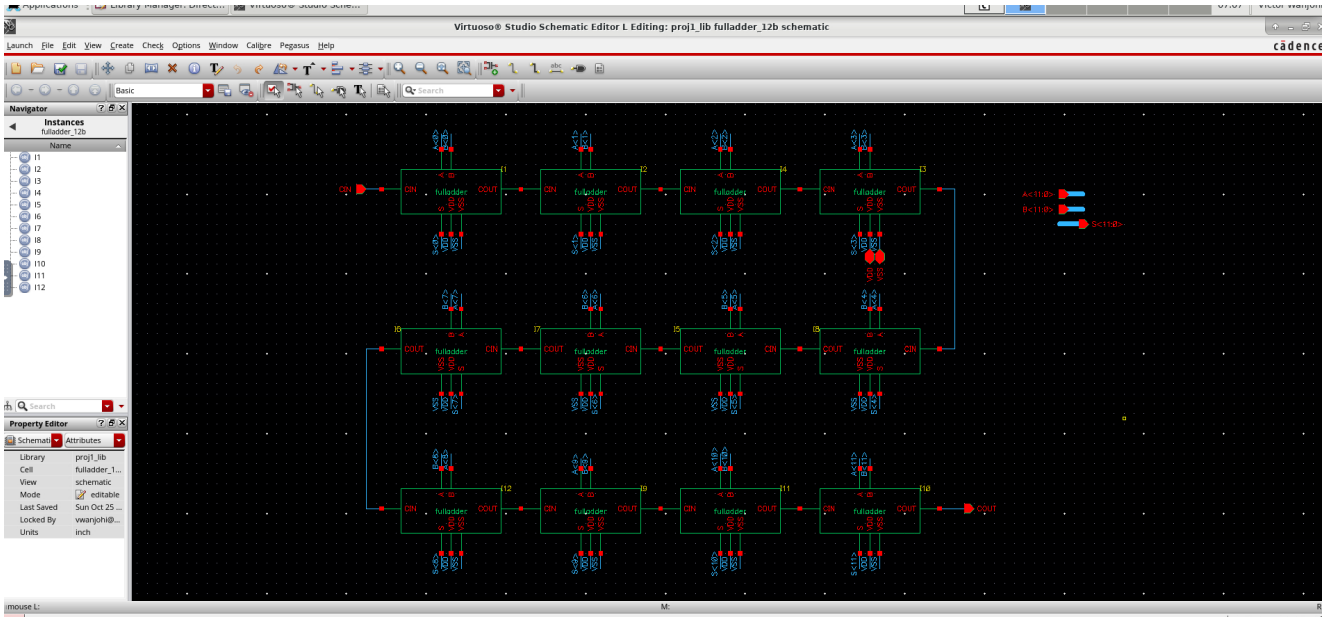


Figure 10: `fulladder_12b`: twelve `fulladder` cells in a ripple chain (I1 bit 0 with C_{in} to I10 bit 11 with C_{out}), drawn as three rows of four; each cell is 148 fins (two `xor2`, two `and2`, one `or2`).

Logical-effort analysis (before simulation). **[TODO: Path from the multiplier input (or register output) through the multiplier and adder to the accumulator register D pin.]**

Stage	Gate	g	p	Size / h
[TODO: 1]				
[TODO: . . .]				

Path effort $F = GBH =$ **[TODO:]**, $N =$ **[TODO:]**, $\hat{f} = F^{1/N} =$ **[TODO:]**, $D = N\hat{f} + P =$ **[TODO:]** τ, $\tau =$ **[TODO: ps]** (FO4 from

Unit tests and predicted vs. measured.

Block	Bench	Test
Multiplier	`tb_multiplier`	Exhaustive: all 256 input pairs, product compared to $a \cdot b$ (error count must be 0); 0 errors for t_v
12-bit accumulator	`tb_accu_12b`	**[TODO: accumulate, clear, hold; carry across all 12 bits (e.g. 0x0FF+1)]**

Multiplier bench (`tb_multiplier4b`, Fig. below). Eight `vpulse` sources count through all 256 (A, B) pairs, one per t_v (A the low nibble: bit k has delay $2^k t_v$, width $2^k t_v$, period $2^{k+1} t_v$; B bit k uses $2^{k+4} t_v$), 20 ps edges, tran stop $256\, t_v$; each product bit drives one `inv_2x` (`LD<7:0>`). **[TODO: Inputs are ideal sources; add two `inv_u` buffers per input or state this.]**

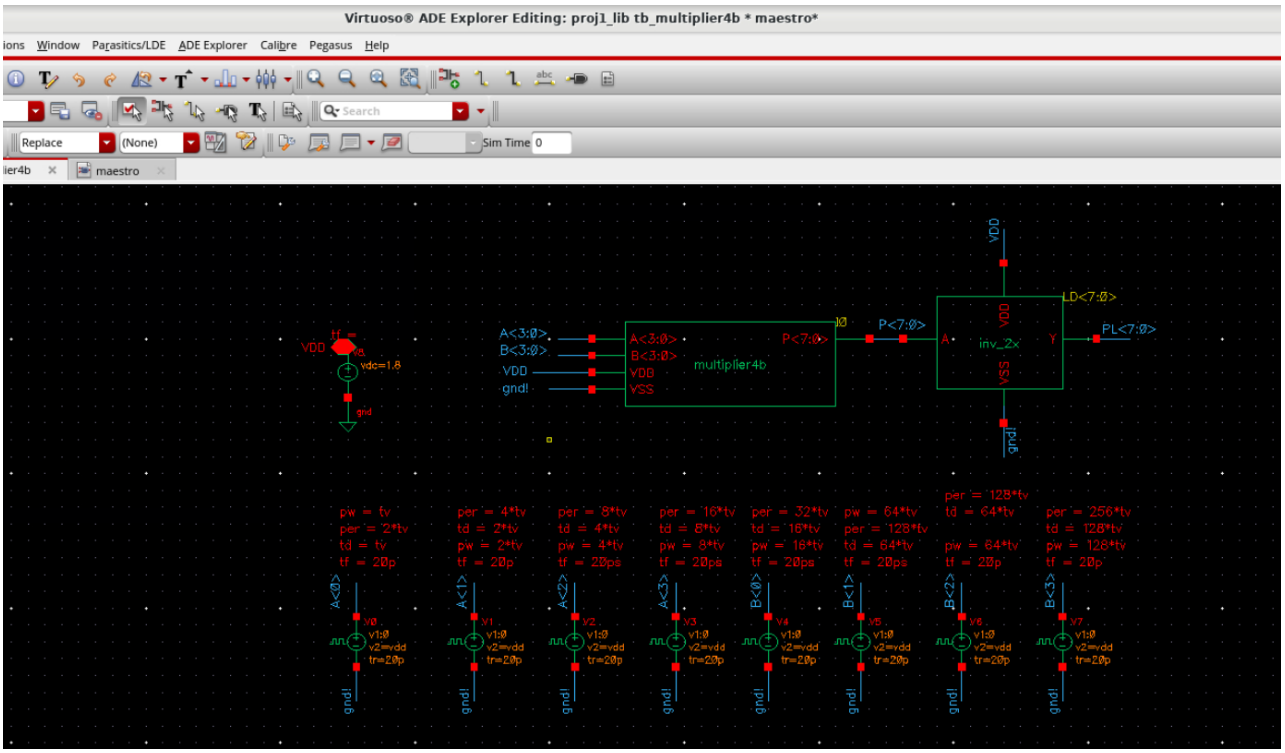


Figure 11: `tb_multiplier4b`. **[TODO: Re-capture with V8 = vdd = 0.8 V (this capture shows vdc = 1.8) and V5 corrected to delay/width $32\,t_v$, period $64\,t_v$.]**

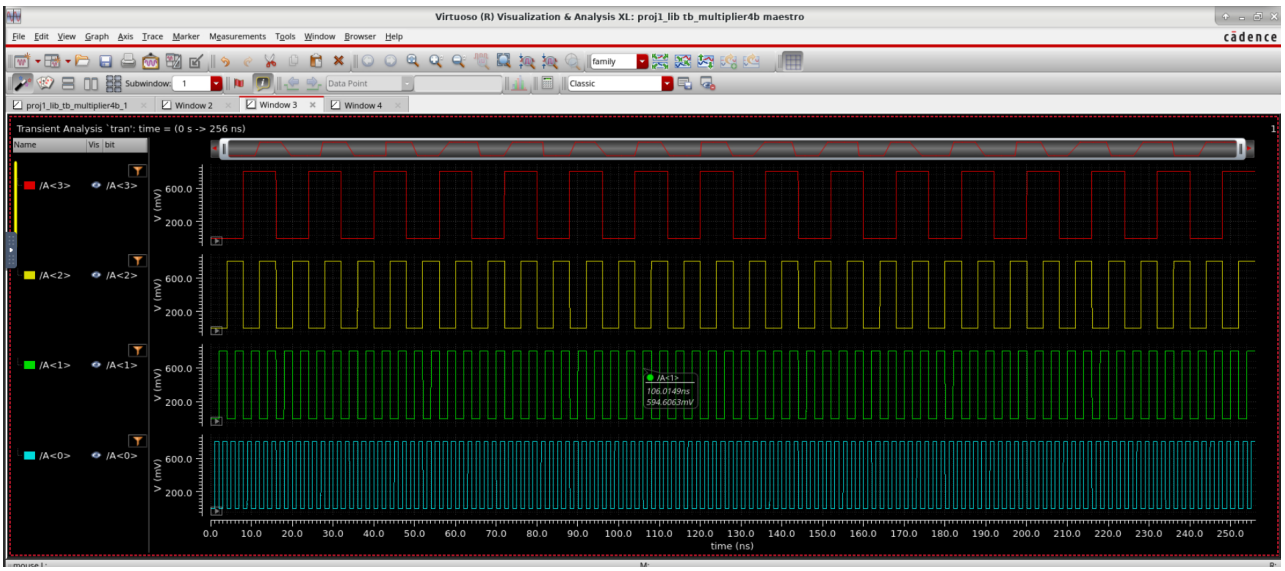


Figure 12: Stimulus $A\langle 3{:}0\rangle$ at $t_v = 1$ ns: A_0 toggles every vector, A_3 every 8.

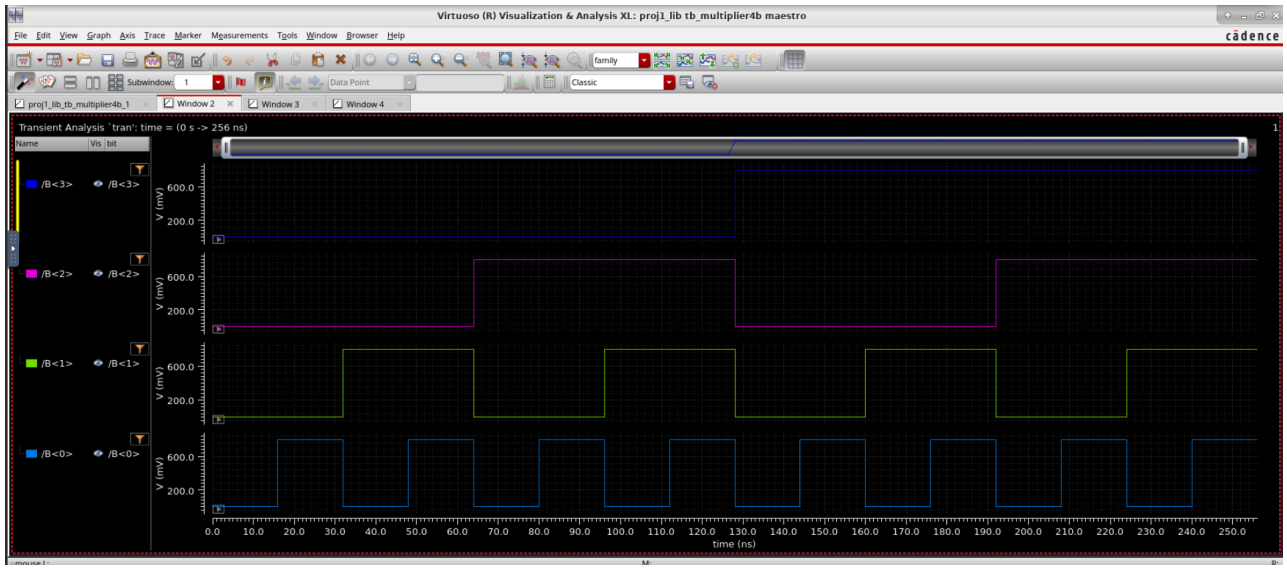


Figure 13: Stimulus $B\langle 3{:}0\rangle$: B steps once every 16 vectors, so all 256 pairs occur in 256 ns.

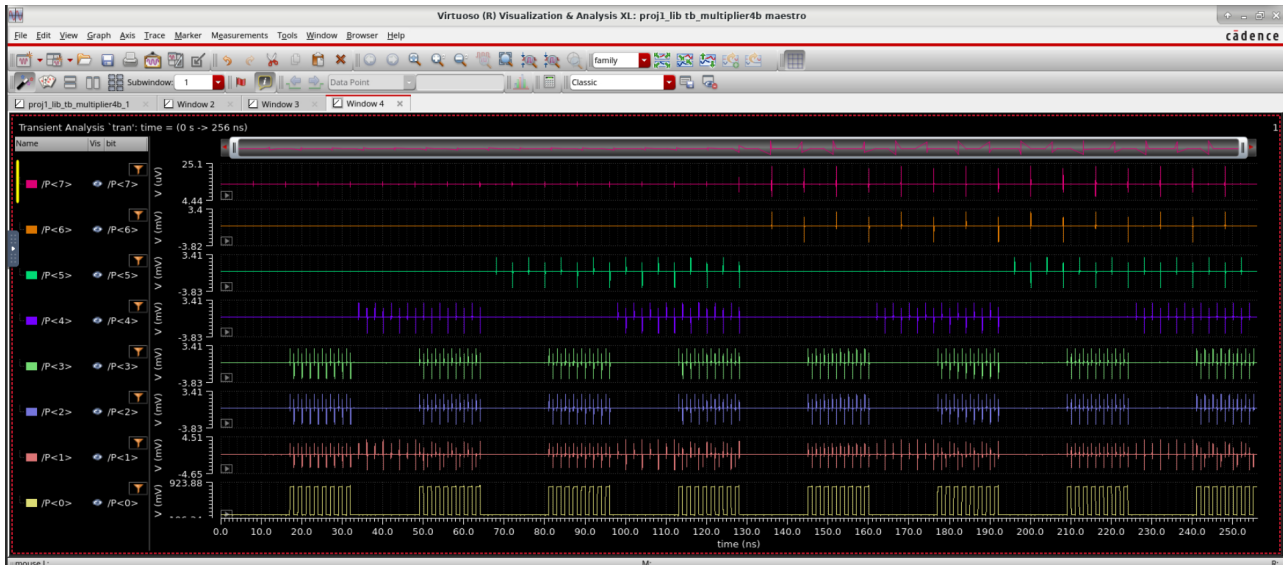


Figure 14: First run (before the fixes): only $P_0 = a_0 b_0$ switches; $P_{1..7}$ stay near 0 V (mV-scale glitches). This located two design errors, below.

Errors found by the first run. (1) `xor2`: the pull-down had the $A/\bar{A}$ devices in parallel, in series with the $B/\bar{B}$ pair in parallel, so it always conducted and fought the pull-up when $A \neq B$; corrected to two series stacks $A \cdot B$ and $\bar{A} \cdot \bar{B}$. (2) `multiplier4b`: all four `and2x4` received $(A\langle 0..3\rangle, B\langle 0..3\rangle)$ and formed $a_i b_i$; each row j now receives B_j on all four B inputs. A logic-level check of the corrected netlist gives 0 wrong products out of 256. **[TODO: Rerun: error count over 256 vectors, worst-case settling time from the t_v sweep, delay of the slowest pair; replace the figure above with the passing run.]**

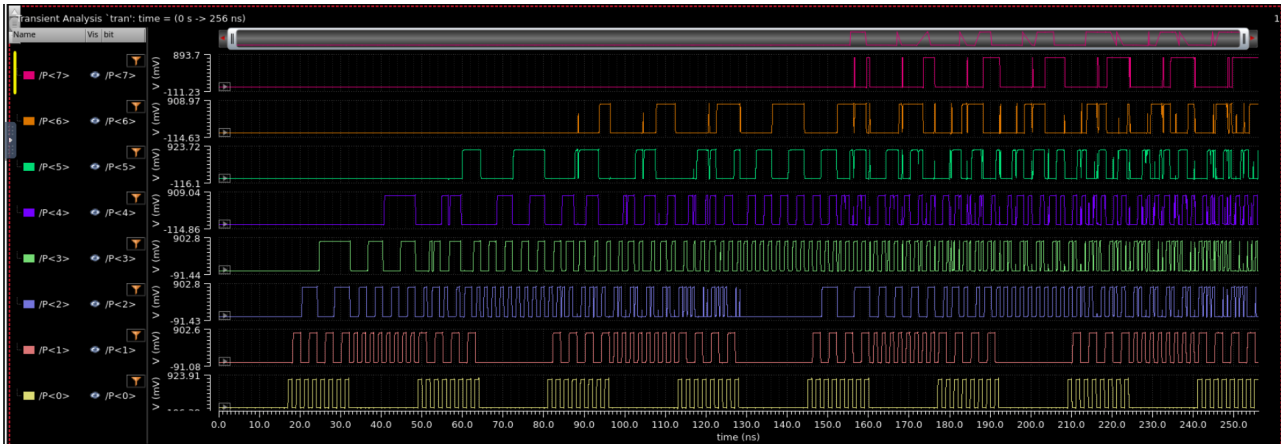


Figure 15: After the fixes ($t_v = 1$ ns): every product bit switches. First rising edges match the hand values: P_1 at vector 18 (2×1), P_3 at 24 (8×1), P_4 at 40 (8×2), P_5 at 59 (11×3), P_7 at 159 ($15 \times 9 = 135$); short spikes are ripple glitches inside a vector. **[TODO: Zoom on the worst-case transition.]**

Exhaustive check. Three ADE expressions turn the bits into integers: $P_{int} = \sum_k 2^k V(P_k)/V_{DD}$, $AB_{int} = (\sum_k 2^k V(A_k)/V_{DD})(\sum_k 2^k V(B_k)/V_{DD})$, and $\text{err} = |P_{int} - AB_{int}|$. `errS` samples err once per vector at $0.95\,t_v$ (`sample(err, 0.95 tv, 255.95 tv, "linear", tv)`) and `errmax` = `ymax(errS)` with spec < 0.5.

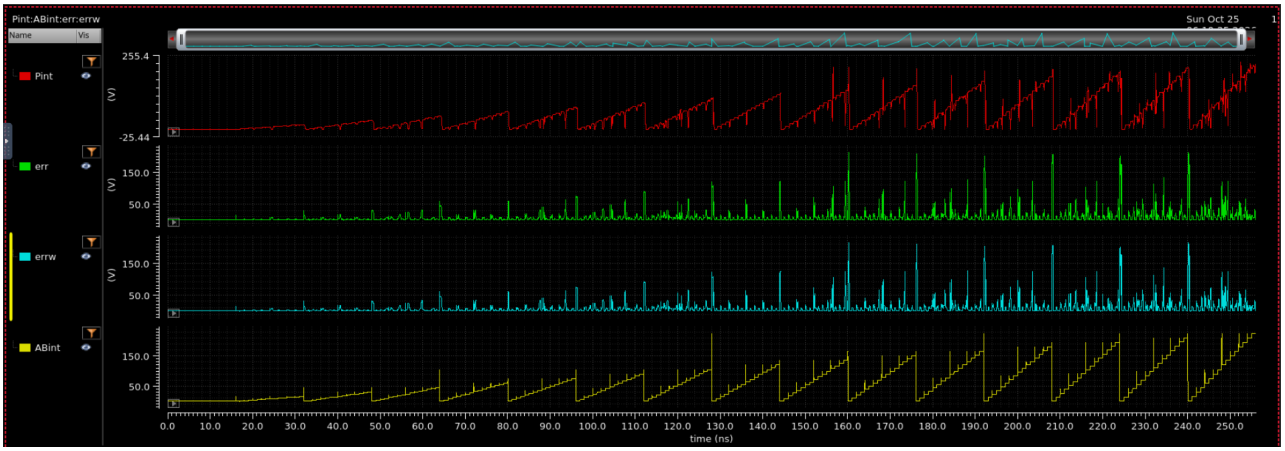


Figure 16: P_{int} follows the AB_{int} staircase in every block of 16 vectors; the err spikes sit at the vector boundaries, where AB_{int} itself spikes because the input bits do not cross at the same instant.

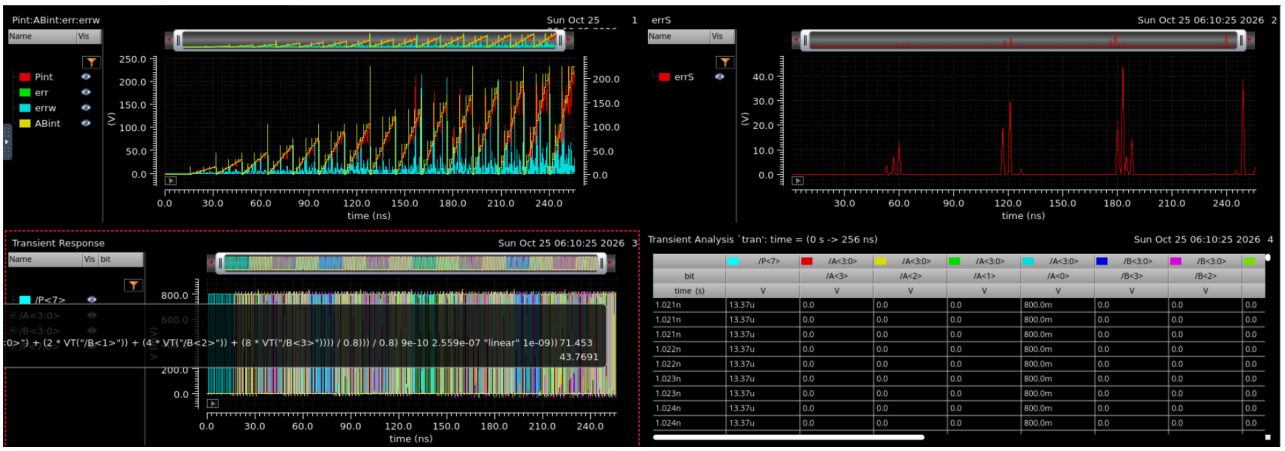


Figure 17: Sampled check at $t_v = 1$ ns (top right, `errS`): 0 for almost all 256 vectors; about ten vectors in four clusters (56–60, 118–122, 180–185, 242–246 ns) are non-zero, peak `errmax` = 43.77.

At $t_v = 1$ ns `errmax` = 43.77 (71.45 with the earlier 0.9 t_v sample point). The non-zero values are fractional, not the power-of-two error a settled wrong bit would give, so those vectors are still switching about 0.95 ns after the inputs: a settling limit, not a logic error.

t_v **sweep.** Tran stop 256 t_v, so every point simulates all 256 vectors; `t_worst` gives the worst vector, $i = \text{round}(t/t_v - 0.95)$.

t_v	sample point ($0.95\,t_v$)	`errmax`	`t_worst`	result (< 0.5)
1.0 ns	0.950 ns	43.77	183.0 ns	fail (vector 182)
1.1 ns	1.045 ns	13.66	269.4 ns	fail (vector 244)
1.2 ns	1.140 ns	0.621	293.9 ns	fail (vector 244)
1.3 ns	**1.235 ns**	**0.170**	323.6 ns	**pass**
1.4 ns	1.330 ns	0.136	348.5 ns	pass
1.5 ns	1.425 ns	0.118	373.4 ns	pass

All 256 products are correct once $t_v \geq 1.3$ ns. The slowest vector is 244 ($B = 15$, A: 0011 $\rightarrow$ 0100, product 45 $\rightarrow$ 60): with $B = 15$ all four partial-product rows are active and three A bits switch, so the carry ripples through the whole array into P_4. It is still settling at 1.14 ns and settled by 1.235 ns, so the **measured worst-case multiplier delay is 1.14–1.24 ns (≈ 1.2 ns)**. A second data point: vector 182 (A 5 $\rightarrow$ 6, $B = 11$) has P_6 crossing $V_{DD}/2$ 962 ps after the input edge (`d_mult` at $t_v = 1$ ns). Typical vectors settle in 0.3–0.6 ns (zoom below). The residual 0.12–0.17 at passing t_v is the bit voltages sitting a few mV off the rails, weighted up to 128.



78 rows

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Parameters: tv=1n						
1	proj1_lib_tb_mul...	/P<7>				
1	proj1_lib_tb_mul...	/A<3:0>				
1	proj1_lib_tb_mul...	/B<3:0>				
1	proj1_lib_tb_mul...	/P<7:0>				
1	proj1_lib_tb_mul...	errmax_old	71.45			
1	proj1_lib_tb_mul...	Pint				
1	proj1_lib_tb_mul...	ABint				
1	proj1_lib_tb_mul...	err				
1	proj1_lib_tb_mul...	errw				
1	proj1_lib_tb_mul...	errS				
1	proj1_lib_tb_mul...	errmax	43.77	< 0.5		fail
1	proj1_lib_tb_mul...	tworst	183n	< (256 * tv)		error
1	proj1_lib_tb_mul...	d_mult	962.1p			
Parameters: tv=1.1n						
2	proj1_lib_tb_mul...	/P<7>				
2	proj1_lib_tb_mul...	/A<3:0>				
2	proj1_lib_tb_mul...	/B<3:0>				
2	proj1_lib_tb_mul...	/P<7:0>				
2	proj1_lib_tb_mul...	errmax_old	206			
2	proj1_lib_tb_mul...	Pint				
2	proj1_lib_tb_mul...	ABint				
2	proj1_lib_tb_mul...	err				
2	proj1_lib_tb_mul...	errw				
2	proj1_lib_tb_mul...	errS				
2	proj1_lib_tb_mul...	errmax	13.66	< 0.5		fail
2	proj1_lib_tb_mul...	tworst	269.4n	< (256 * tv)		error
2	proj1_lib_tb_mul...	d_mult	eval err			
Parameters: tv=1.2n						
3	proj1_lib_tb_mul...	/P<7>				
3	proj1_lib_tb_mul...	/A<3:0>				
3	proj1_lib_tb_mul...	/B<3:0>				
3	proj1_lib_tb_mul...	/P<7:0>				
3	proj1_lib_tb_mul...	errmax_old	158.9			
3	proj1_lib_tb_mul...	Pint				
3	proj1_lib_tb_mul...	ABint				
3	proj1_lib_tb_mul...	err				
3	proj1_lib_tb_mul...	errw				
3	proj1_lib_tb_mul...	errS				
3	proj1_lib_tb_mul...	errmax	621m	< 0.5		fail
3	proj1_lib_tb_mul...	tworst	293.9n	< (256 * tv)		error
3	proj1_lib_tb_mul...	d_mult	669.2p			
Parameters: tv=1.3n						
4	proj1_lib_tb_mul...	/P<7>				
4	proj1_lib_tb_mul...	/A<3:0>				
4	proj1_lib_tb_mul...	/B<3:0>				
4	proj1_lib_tb_mul...	/P<7:0>				
4	proj1_lib_tb_mul...	errmax_old	164.2			
4	proj1_lib_tb_mul...	Pint				
4	proj1_lib_tb_mul...	ABint				
4	proj1_lib_tb_mul...	err				
4	proj1_lib_tb_mul...	errw				
4	proj1_lib_tb_mul...	errS				
4	proj1_lib_tb_mul...	errmax	170.3m	< 0.5		pass
4	proj1_lib_tb_mul...	tworst	323.6n	< (256 * tv)		error
4	proj1_lib_tb_mul...	d_mult	eval err			
Parameters: tv=1.4n						
5	proj1_lib_tb_mul...	/P<7>				
5	proj1_lib_tb_mul...	/A<3:0>				
5	proj1_lib_tb_mul...	/B<3:0>				
5	proj1_lib_tb_mul...	/P<7:0>				
5	proj1_lib_tb_mul...	errmax_old	89.95			
5	proj1_lib_tb_mul...	Pint				
5	proj1_lib_tb_mul...	ABint				
5	proj1_lib_tb_mul...	err				
5	proj1_lib_tb_mul...	errw				
5	proj1_lib_tb_mul...	errS				
5	proj1_lib_tb_mul...	errmax	136.2m	< 0.5		pass
5	proj1_lib_tb_mul...	tworst	348.5n	< (256 * tv)		error
5	proj1_lib_tb_mul...	d_mult	eval err			
Parameters: tv=1.5n						
6	proj1_lib_tb_mul...	/P<7>				
6	proj1_lib_tb_mul...	/A<3:0>				
6	proj1_lib_tb_mul...	/B<3:0>				
6	proj1_lib_tb_mul...	/P<7:0>				
6	proj1_lib_tb_mul...	errmax_old	63.12			
6	proj1_lib_tb_mul...	Pint				
6	proj1_lib_tb_mul...	ABint				
6	proj1_lib_tb_mul...	err				
6	proj1_lib_tb_mul...	errw				
6	proj1_lib_tb_mul...	errS				
6	proj1_lib_tb_mul...	errmax	118.4m	< 0.5		pass
6	proj1_lib_tb_mul...	tworst	373.4n	< (256 * tv)		error
6	proj1_lib_tb_mul...	d_mult	eval err			

Figure 18: ADE Results for the sweep: `errmax` (spec < 0.5) fails at 1.0–1.2 ns and passes from 1.3 ns. (The "error" on `t_worst` is only its spec, which cannot use a design variable; `d_mult` uses a fixed 182–183 ns window and is valid only at $t_v = 1$ ns; `errmax_old` uses fixed sample times.)

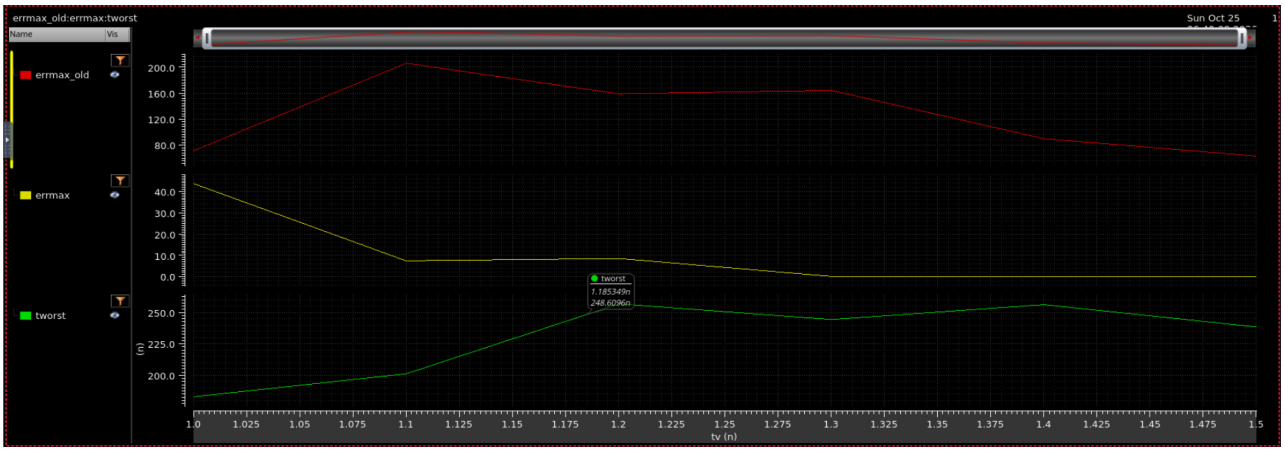


Figure 19: `errmax` and `t_worst` against t_v from an earlier run with a fixed 256 ns stop (covers fewer vectors above 1 ns); kept for the shape of the curve, the table above is the result.

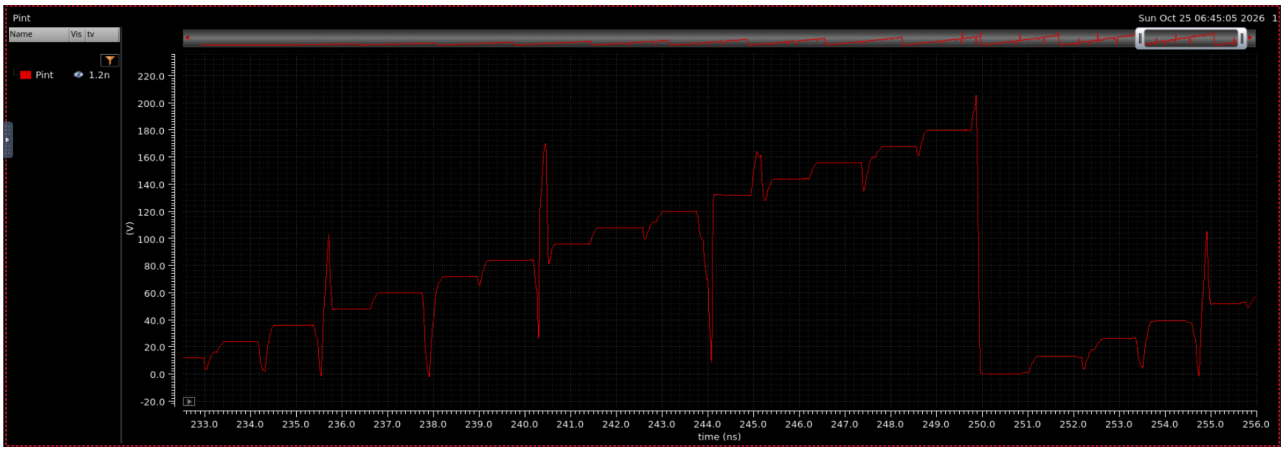


Figure 20: P_{int} at $t_v = 1.2$ ns, 233–256 ns ($B = 12$, then $B = 13$): each product settles 0.3–0.6 ns after its vector edge; the spike near 249.9 ns is the 180 → 0 step where all four A bits and B_0 flip.

[TODO: Zoom on vector 244 at $t_v = 1.3$ ns (317.2–318.5 ns): which P bit settles last, input-edge-to-0.4 V delay (the measured critical path for the logical-effort comparison).]

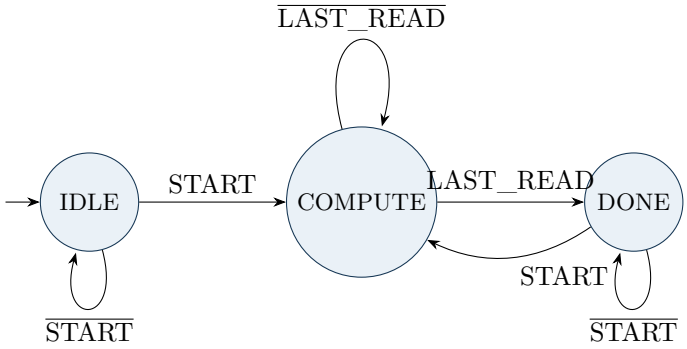
[missing figure: tb_accu_12b_tran]

Figure 21: **[TODO: Accumulator test, including the longest carry propagation.]**

4. Sequencer (3.3)

Rubric (3.3): 4-bit address counter and control FSM (idle, compute, done) built from the registers; state diagram; timing of START, counter, memory reads, accumulator update and DONE over one dot product.

State diagram.



Encoding (from `controller_mac`): two state registers, S_0 and DONE; IDLE $= (S_0, \text{DONE}) = (0, 0)$, COMPUTE: RUN $= S_0\overline{\text{DONE}} = 1$, DONE $= (0, 1)$. Two more registers carry the pipeline: VALID (= RUN one cycle late, so the first product is accumulated one cycle after its read) and LAST_READ (set the cycle after the counter reaches 15). RST clears S_0 and DONE from any state. Per state:

- **IDLE / DONE, START high:** $G_0 = \text{START} \cdot \overline{\text{RUN}}$ raises CLEAR (clears the accumulator, the counter, VALID and LAST_READ) and sets S_0; DONE clears. START is ignored while RUN.
- **COMPUTE:** COUNT_EN $= \text{RUN} \cdot \overline{\text{LAST}}$ steps the counter 0 to 15 and holds it at 15 (LAST = count = 15); ACC_EN $= \text{RUN} \cdot \text{VALID}$; S_0 stays set while LAST_READ $= 0$.
- **Leaving COMPUTE:** FINISH $= \text{RUN} \cdot \text{LAST_READ}$ sets DONE and clears S_0, so the exit is one cycle after the count reaches 15, which leaves time to accumulate the last product.
- **DONE:** held by DONE $\cdot \overline{\text{START}}$; ACC holds because ACC_EN $= 0$.

Counter and controller. `counter_4b` is a synchronous toggle-chain counter: $T_1 = EN \cdot Q_0$, $T_2 = T_1 Q_1$, $T_3 = T_2 Q_2$ (`and2`), $D_k = (Q_k \oplus T_k) \cdot \overline{RST}$ with $T_0 = EN$ (`xor2`, `and2`), four `d_ff`; RST is driven by CLEAR, so the count restarts at 0 on START (728 fins). `controller_mac` is the logic above: 4 `register_1b`, 11 `and2`, 3 `or2`, 5 `inv_2x` (948 fins). In `top`, four `mux2` send COUNT to the memory address while RUN = 1 and the external ADDR otherwise, and LOAD_EN $= \overline{\text{RUN} + \text{DONE} + \text{START}}$ gates WE_W and WE_X. **[TODO: `tb_counter4b` and `tb_controller` results (count 0 to 15 and wrap, hold, reset; START during compute ignored, DONE hold, restart from DONE).]**



Figure 22: `counter_4b`: four `d_ff` (I5–I8); $D_k = (Q_k \oplus T_k) \cdot \overline{RST}$ through `xor2` and `and2` (I12/I13, I21/I22, I25/I26, I24/I23); toggle chain $T_1 = EN \cdot Q_0$, $T_2 = T_1 Q_1$, $T_3 = T_2 Q_2$ (I15, I27, I28); RST inverted by I20. 728 fins.

[missing figure: controller_schematic]

Figure 23: **[TODO: Controller.]**

[missing figure: tb_counter4b_tran]

Figure 24: **[TODO: Counter waveform.]**

[missing figure: tb_controller_tran]

Figure 25: **[TODO: Controller waveform.]**

Timing over one dot product. **[TODO: Annotated diagram, as in Part 2 (`timing_diagram_2p2c.tex` is the model): CLK, START, state, counter, ADDR to the macros, W/X outputs, product, ACC, DONE; mark the START edge, the first read, the first accumulate, the last accumulate, DONE rising.]**

Architecture	pipelined: address and read in cycle n, multiply-accumulate in $n+1$ (VALID)
Cycles START to DONE	17 edges after the START edge (cycle-level model of the netlist; limit 20) **[TODO: confirm in `tb_top`]**
Cycles per dot product	17 (plus 32 load cycles when the memories are reloaded)

5. Timing closure (3.4)

Rubric (3.4): critical path identified; minimum period by hand from $t_{CLK\to Q}$/SRAM access, the logical-effort datapath estimate, and register t_{cq} and t_{su}; measured as the shortest passing period; hold checked on fast paths.

Critical path. The pipeline has no register between the macros and the multiplier: the macro's own address latch captures COUNT at the rising edge and its output latch holds DOUT for the rest of the cycle. The paths that must fit in one period are therefore (1) counter t_{cq} → `mux2` → macro address latch (short), and (2) CLK edge → SRAM access → multiplier → 12-bit adder → accumulator setup, the critical path: $T_{\min} \geq t_{\text{SRAM}} + t_{\text{mult}} + t_{\text{add}} + t_{su}$, with t_{SRAM} measured from CLK and so already including the macro's internal address latch.

$$T_{\min} \geq t_{cq}+t_{\text{SRAM}}+t_{\text{mult}}+t_{\text{add}}+t_{su}+t_{\text{skew}} \quad \text{(non-pipelined)}; \qquad T_{\min} \geq \max\left(t_{cq}+t_{\text{SRAM}}+t_{su},\ t_{cq}+t_{\text{mult}}+t_{\text{add}}+t_{su}\right) \quad \text{(pipelined)} \tag{1}$$

Term	Value	Source
t_{cq} (register)	**[TODO: ps]**	§2
t_{SRAM} (CLK to DOUT)	488 ps	Part 2, worst case at $T_{CLK} = 0.69$ ns
$t_{\text{mult}} + t_{\text{add}}$	$t_{\text{mult}} \approx 1200$ ps measured (1140–1240); **[TODO: LE estimate, t_{add}]**	§3
t_{su}	**[TODO: ps]**	§2
Hand $T_{\min}$	**[TODO: ns]**	sum
Measured $T_{\min}$	**[TODO: ns]**	shortest period at which tests A, B, C

Period sweep. **[TODO: Table as in Part 2 §6: t_{clk}, ACC error for A/B/C, result.]** Discuss the gap between hand and measured value: **[TODO:]**.

[missing figure: tb_top_tclk_sweep]

Figure 26: **[TODO: Pass/fail against clock period.]**

Hold check at the fast paths. Two short paths: (1) counter → `mux2` → macro address latch, which closes at the same rising edge that changes the counter, so the new count must arrive after the latch has closed: $t_{cq,\min}+t_{\text{mux}} \geq t_{h,\text{latch}}$; (2) register to register in the controller and counter (e.g. VALID from RUN through one `and2`): $t_{cq,\min}+t_{\text{logic},\min} \geq t_h+t_{\text{skew}}$. **[TODO: Margins in ps from `tb_dff` ($t_{cq,\min}$, t_h) and a simulation of path (1) in `tb_top`.]**

6. Verification (3.5)

Rubric (3.5): three test vectors; ACC and DONE waveforms for Test A; pass/fail table for all three.

Bench (`tb_top`). **[TODO: Load phase: START low, WE_W/WE_X write 16 words each through ADDR/DIN, one per cycle; compute phase: START pulse, wait for DONE, sample ACC. Inputs through realistic buffers as in Part 2. ACC compared to the expected value with an error measure.]**

Test	$W[0..15]$	$X[0..15]$	Expected ACC
A	3, 12, 7, 0, 15, 9, 1, 14, 6, 11, 2, 13, 8, 5, 10, 4	9, 2, 15, 7, 4, 12, 0, 11, 13, 1, 8, 6, 3, 14, 5, 10	845 (0x34D)
B	all 15	all 15	3600 (0xE10)
C	5, 10, 5, 10, ...	10, 5, 10, 5, ...	800 (0x320)

Test	Expected	ACC read	DONE high at cycle	Cycles	Result
A	845 (0x34D)	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO: pass/fail]**
B	3600 (0xE10)	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**
C	800 (0x320)	**[TODO:]**	**[TODO:]**	**[TODO:]**	**[TODO:]**

Test B is the full-scale case (largest carry chains, no overflow); Test C alternates 5×10 and 10×5 to maximize toggling on the memory outputs and multiplier inputs.

[missing figure: tb_top_testA_acc_done]

Figure 27: **[TODO: Test A: ACC⟨11:0⟩ as a bus and DONE; mark START, first accumulate, DONE rising, final value 0x34D, and that ACC holds until the next START.]**

[missing figure: tb_top_testA_partial_sums]

Figure 28: **[TODO: Optional: running partial sums against the hand-computed values for Test A.]**

[TODO: Also check, if time allows: RST mid-computation, START during compute, back-to-back dot products without reloading (ACC clears on START).]

7. Metrics and figure of merit

Rubric (handout): $T_{CLK,\min}$ and $t_{dot} = N_{cycles} \cdot T_{CLK,\min}$ (START to DONE); E_{dot} = supply energy from START to DONE at $T_{CLK,\min}$ for Test A, also B and C, split into memory, datapath, sequencer if separate supplies; area = 2 macros (Part 2) + datapath and sequencer with D; FOM_{MAC} = Area $\cdot E_{dot} \cdot t_{dot}$ (μm^2, pJ, ns); energy per MAC with and without the memory accesses; paragraph on where the energy goes.

At $V_{DD} = 0.8$ V, nominal, 27°C, $D = 28$:

Metric	Value
$T_{CLK,\min}$	**[TODO: ns]** (measured); hand estimate **[TODO: ns]**
N_{cycles} (START to DONE)	17 (predicted) **[TODO: confirm]**
$t_{dot} = N_{cycles}\, T_{CLK,\min}$	**[TODO: ns]**
Area: 2 macros + datapath + sequencer	2×54.6 *(est.)* $+ D \sum W_i L_i$ (7544 fins $\times$ 0.0242 μm^2: datapath 5696 fins = 137.8, sequencer and glue 1848 fins = 44.7) = **291.8 μm^2** *(est.)*
FOM$_{MAC}$ = **Area**$\cdot E_{dot} \cdot t_{dot}$ (Test A)	**[TODO: μm^2·pJ·ns]**

Energy (integral of supply current $\times V_{DD}$ from START to DONE, at $T_{CLK,\min}$):

Test	E_{dot} total	Memory (2 macros)	Datapath	Sequencer
A	**[TODO: pJ]**	**[TODO:]**	**[TODO:]**	**[TODO:]**
B (all 15)	**[TODO: pJ]**	**[TODO:]**	**[TODO:]**	**[TODO:]**
C (5/10 alternating)	**[TODO: pJ]**	**[TODO:]**	**[TODO:]**	**[TODO:]**

[TODO: State how the split is obtained (separate supply sources per block, with the registers counted in the block they belong to).]

	With memory accesses	Without memory accesses
Energy per MAC, Test A (E_{dot}/16)	**[TODO: pJ]**	**[TODO: pJ]**

Where the energy goes. **[TODO: Paragraph: share of memory against datapath and sequencer; clock tree and register energy per clock (§2) times the number of registers; why B and C differ from A (toggling); static versus dynamic.]**

8. What limited the design

The clock period is set by the multiply-accumulate path, not the memory: the array multiplier alone takes about 1.2 ns in its worst case, over twice the 488 ps SRAM read, before the 12-bit ripple adder and the register setup are added. Both arithmetic blocks are ripple structures built from one small full adder (2-fin `xor2` stacks with input inverters), which made them easy to verify but long. In area the datapath (138 μm^2) is larger than the two macros (109 μm^2 *(est.)*), mostly the 24 full adders and the 12 accumulator registers. **[TODO: Energy split once measured.]** With more time: a carry-save multiplier, a faster final adder (carry-select or lookahead), upsized `xor2` stacks, and a pipeline register after the multiplier.

- **Register style:** static TG master-slave, 100 fins per flip-flop (152 with enable and reset), reused everywhere; a dynamic latch would be smaller and faster but cannot hold with the clock stopped. **[TODO: t_{cq}, t_{su}, energy per clock.]**
- **Pipelined vs. not:** overlapping the read with the multiply-accumulate costs one cycle (17 instead of 16) and stays within the 20-cycle limit; because the macro output latch is the only pipeline stage, the SRAM access and the arithmetic still share one period. A register after the multiplier would split them, at a second extra cycle (18).
- **Multiplier and adder architecture:** ripple array and ripple adder: smallest design effort and regular, but the slowest choice; a carry-save tree and a carry-select final adder are the first changes for speed.